Cairo University - Faculty Of Engineering
Computer Engineering Department
CMP4007 Applied Data Science - Spring 2026

Project Final Report

Team 9

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1 Problem Definition & Business Context

1.1 Business Problem

Used car dealerships and online marketplaces receive large volumes of vehicle listings and trade-ins. Manually assigning each vehicle into a pricing/value tier is time-consuming and inconsistent across staff and branches. This project aims to automate the classification of incoming used cars into value tiers using historical market data.

1.2 Stakeholders

- Primary: Used Car Dealerships (pricing and inventory teams)
- Secondary: Online Marketplaces (listing recommendations) and Insurance Underwriters (risk and valuation support)

1.3 Problem Formulation

This problem is formulated as a **classification task**, where the goal is to predict the **price/value tier** of a used car listing.

Tier Definition: Cars are grouped into **Budget**, **Mid-Range**, and **Luxury** tiers based on price.

1.4 Justification for Classification

- The target is a discrete set of tiers (Budget / Mid-Range / Luxury)
- Tier outputs support operational decisions (pricing, procurement, and marketing)
- Classification enables threshold-based policies and consistent segmentation

1.5 Business Impact

- Faster and more consistent inventory categorization
- Improved pricing strategy and margin optimization
- Better customer targeting (budget buyers vs. premium buyers)

2 Data Acquisition & Integration

2.1 Sources Description

2.1.1 Primary Source (Kaggle Dataset)

Source: Kaggle — *Uncovering Factors That Affect Used Car Prices*
URL: https://www.kaggle.com/datasets/thedevastator/uncovering-factors-that-affect-used-car-prices

autos.csv

Main structured dataset (CSV). Each row represents a used car listing with attributes such as price, brand, model, registration year, mileage, and fuel type.

2.1.2 Secondary Source (Web Scraping)

Source: AutoScout24
URL: `https://www.autoscout24.com`

AutoScout24 Listings

Scraped used car listings to increase coverage and validate pricing patterns.

2.2 Data Acquisition

- Kaggle dataset downloaded as CSV file(s) (e.g., `autos.csv`).
- AutoScout24 listings collected via web scraping using Python (e.g., `Scrapy` + `Scrapy-Splash`).

2.3 Data Integration Strategy

- **Vertical merge** the Kaggle dataset with scraped AutoScout24 listings by aligning both sources to a unified schema.
- Standardize fields and resolve naming differences (e.g., `kilometers` vs `mileage`).
- Deduplicate near-identical listings using matching keys such as **brand**, **model**, **kilometers**, **yearOfRegistration**, and **power**.
- Merge strategy is union rows after schema alignment.
- Handle missing fields introduced by scraping or schema differences using imputation or "unknown" categories.

Challenges:

- Inconsistent naming and category sets across sources
- Translating the Kaggle dataset from German to English
- Missing or noisy scraped fields due to unstructured HTML
- Potential duplicates across sources

2.4 Feature Documentation (Core Fields)

The accepted features to be used after the merging and removal of redundant features:

#	Feature	Type	Description
1	seller	Categorical	seller type (business/private)
2	vehicleType	Categorical	body type
3	yearOfRegistration	Numeric	Year the car was registered
4	gearbox	Categorical	Type of gearbox (manual or automatic)
5	power	Numeric	Power of the car in Horse Power
8	model	Categorical	Model of the car
6	brand	Categorical	Brand of the car
7	kilometers	Numeric	Kilometers the car has been driven
8	fuelType	Categorical	Type of fuel (e.g. diesel, petrol, etc.)
9	dataSource	Categorical	Source of the data (kaggle/crawled)
10	price	Numeric	Price of the car

Note: After deriving the tier label from `price`, the raw `price` feature will not be used as an input feature to avoid target leakage.

3 Data Validation & Documentation

3.1 Dataset Validation Pipeline

Step	Description	Outcome
1	Schema Validation	Data type report
2	Completeness Check	Missing value report
3	Duplicate Detection	Duplicate count
4	Descriptive Statistics	Summary stats
5	Range & Outlier Checks	Violation report
6	Categorical Analysis	Category distribution
7	Distribution Checks	Distribution plots
8	Outlier Analysis	IQR & Z-Score & Isolation Forest Outliers

3.2 Dataset Validation Report

Metric	Value
Total checks	9
Checks Passed	1
Checks FAILed	8
Success Rate	11.1%
Full-row duplicates	57,324 (15.33%)

3.2.1 Completeness - Missing Values Analysis

Column	Missing %	Status
brand	0.00%	PASS
model	5.48%	FAIL (> 5%)
vehicleType	10.12%	FAIL (> 5%)
power	0.00%	PASS
gearbox	5.40%	FAIL (> 5%)
kilometer	0.01%	PASS
fuelType	8.93%	FAIL (> 5%)
yearOfRegistration	0.02%	PASS
seller	0.00%	PASS
price	0.00%	PASS

3.2.2 Descriptive Statistics (Numeric Columns)

Column	Mean	Std	Mode	Min	Q1	Median	Q3	Max
yearOfRegistration	2004.68	92.57	2000	1000	1999	2003	2008	9999
power	115.98	191.77	0	0	70	105	150	20 K
kilometer	125.3 K	40.5 K	150 K	0	100 K	150 K	150 K	999.9 K
price	17.4 K	3.6 M	0	0	1,150	2,990	7,350	214.8 M

3.2.3 Distribution Checks

Column	Skewness	Skewness Label	Kurtosis	Kurtosis Label
yearOfRegistration	2004.68	2003	92.57	72.35
price	17369.87	2990	3575910.38	580.00
kilometer	125298.65	150000	40548.44	-1.48
power	115.98	105	191.77	58.14

3.2.4 Consistency - Schema & Data Types

Column	Detected Type	Expected Type	Match
yearOfRegistration	float64	int64	No
model	str	str	Yes
power	float64	float64	Yes
kilometer	float64	float64	Yes
price	int64	int64	Yes
seller	str	str	Yes
gearbox	str	str	Yes
vehicleType	str	str	Yes
fuelType	str	str	Yes
brand	str	str	Yes

3.2.5 Uniqueness - Duplicate Detection

- Full-row duplicates detected: **57,324 rows (15.33%)**

3.2.6 Validity Checks

Column	Rule	Violations	Status
yearOfRegistration	[1900 - 2026]	182 (68 below, 114 above)	FAIL
kilometer	[0 - 300,000]	13 (above max)	FAIL
power	[5 - 5,000]	40,988 (40,903 below, 85 above)	FAIL
price	[500 - 5,000,000]	36,114 (36,062 below, 52 above)	FAIL

3.2.7 Accuracy - Range & Outlier Checks

Column	Rule	Violations	Status
yearOfRegistration	[1950 - 2026]	182 (68 below, 114 above)	FAIL
kilometer	[0 - 299,999]	13 (above max)	FAIL
power	[1 - 4,999]	40,907 (40,820 below, 87 above)	FAIL
price	[0 - 1,000,000]	65 (above max)	FAIL

3.2.8 Statistical Outlier Detection

Column	Z-Score Outliers ($\|z\| > 3.0$)	Z-Score %	IQR Outliers	IQR %	Status %
price	40	0.01%	28,424	7.60%	FAIL
power	385	0.10%	11,035	2.95%	FAIL
kilometer	290	0.08%	15,397	4.12%	FAIL
yearOfRegistration	173	0.05%	8,289	2.22%	FAIL

Multivariate Outlier Detection (Isolation Forest):

- Detected **44,875 outliers (12.00%)** across combined numeric features (`price`, `power`, `kilometer`, `yearOfRegistration`).

3.3 Quality Issues Summary

- **Schema Issue:** `yearOfRegistration` is formatted as a `float64` instead of the expected `int64`.
- **Missing Data:** Significant missing values across `vehicleType` (10.12%), `fuelType` (8.93%), `model` (5.48%), and `gearbox` (5.40%).
- **Duplicate Records:** High proportion of exact full-row duplicates detected (15.33%).
- **Invalid Ranges:** Massive positive skew in `price` (max 2B+) and `power` (max 20k) indicating corrupted entries. `power` heavily violates minimum range boundaries (40,903 values below 5).
- **Isolation Forest:** flagged 44,875 anomalous records (12.00% of the dataset).

3.4 Target Variable Definition

Target Definition

PRICE_TIER = value tier derived from listing `price`.

Tier Construction:

Tier	Label	Price Range	Business Meaning	Business Interpretation
0	Budget	$\leq$ \$3,500	High-mileage or older cars	Entry-level pricing segment
1	Mid-Range	\$3,500 - \$12,000	Recent-model mid-size cars	Mainstream segment
2	Luxury	$>$ \$12,000	Luxury, electric, or new cars	Premium segment

Class Distribution [54.81% Budget, 37.19% Mid-Range, 7.99% Luxury]

4 Preprocessing, Transformation & Engineering Data

4.1 Preprocessing Pipeline

Step	Description	Tools	Outcome
1	Data Cleaning	Pandas	Cleaned dataset
2	Outlier Detection	IQR, Z-score	Outlier report
3	Feature Transformation	Scikit-learn	Transformed features
4	Feature Engineering	Custom code	New features
5	Feature Selection	Scikit-learn	Selected features
6	Train-Validation-Test Split	Scikit-learn	Data splits
7	Data Balancing	SMOTE, Undersampling	Balanced training set

4.2 Cleaning Steps & Justification

- Remove invalid rows:
  - year outside [1900, 2026]
  - kilometers outside [0, 300000]
  - price is missing or price < 100
- Keep one record of the duplicated records
- Impute missing categorical values using mode value of the brand
- Standardize categorical labels across sources

4.3 Outlier Detection & Handling

Column	Detection Method	Handling Strategy	Outcome
powerPS	Z-score	Cap to 500	Reduced skewness
price	IQR	Cap to 100,000	Reduced skewness
kilometers	IQR	Cap to 200,000	Reduced skewness

4.4 Feature Transformation& Justification

Column	Transformation	Justification	Outcome
powerPS	Log transformation	Reduce skewness	More normal distribution
price	Log transformation	Reduce skewness	More normal distribution
kilometers	Log transformation	Reduce skewness	More normal distribution

4.5 Feature Engineering

In addition to the core fields, we have engineered the following features:

- **vehicleAge** $=$ current_year (2026) $-$ yearOfRegistration

- **kmPerYear** = kilometers / (vehicleAge + 1)
- **isAutomatic** = (gearbox == "automatic")
- **is_vintage_classic** derived from vehicleAge & brand
- **power_to_tier_bin** derived from powerPS

4.6 Feature Selection & Justification

Feature	Selection Method	Justification	Outcome
vehicleAge	Mutual Information	High relevance to price	Selected
kmPerYear	Mutual Information	Captures usage intensity	Selected
isAutomatic	Chi-Squared Test	Categorical relevance	Selected
is_vintage_classic	Domain Knowledge	Vintage/classic cars have unique pricing	Selected
power_to_tier_bin	Mutual Information	Captures power distribution	Selected

4.7 Train-Validation-Test Split Procedure

Dataset	Split Ratio	Method	Outcome
Training Set	70%	Stratified Random Sampling	260,070 rows
Validation Set	15%	Stratified Random Sampling	55,729 rows
Test Set	15%	Stratified Random Sampling	55,729 rows

4.8 Data Balancing Strategy

Technique	Description	Justifica
SMOTE	Synthetic Minority Over-sampling Technique	Generates synthetic samples f
Undersampling	Randomly removes samples from majority class (Budget)	Reduces dominance

4.9 Preprocessing Summary

- Cleaned dataset by removing invalid rows and handling duplicates
- Detected and handled outliers in power, price, and kilometers
- Applied log transformation to reduce skewness in numeric features
- Engineered new features such as vehicleAge, kmPerYear, isAutomatic, is_vintage_classic, and power_to_tier_bin
- Selected relevant features based on mutual information and domain knowledge
- Split data into stratified train (70%), validation (15%), and test (15%) sets
- Balanced training set using SMOTE and undersampling to address class imbalance

Dataset	Before Preprocessing	After Preprocessing
Total rows	371,528	350,000
Total columns	21	15
Missing values	194,786	0
Outliers	Detected in power, price, kilometers	Handled by capping
Class distribution	54.81% Budget, 37.19% Mid-Range, 7.99% Luxury	Balanced to 33.3% each

5 Exploratory Data Analysis (EDA)

5.1 EDA Dashboard

6 Model Development

7 Experiment Tracking with MLflow

8 Testing

9 Results & Evaluation

10 Code Automation

11 Continous Integration

12 Deployment & Monitoring

13 Interactive Dashboard

14 Limitations & Future Work

14.1 Limitations

We acknowledge the following limitations in our current approach:

- Data quality issues such as missing values and potential outliers may impact model performance.
- Class imbalance, even after balancing techniques, may still affect the model's generlization.
- The model may not capture all relevant features influencing car pricing, such as regional market differences or economic factors.
- The current model is based on historical data and needs retraining on future market changes.

14.2 Future Work

Future work could include:

- Investigating the impact of external factors (e.g., economic conditions, seasonal trends) on car pricing and incorporating them into the model.
- Exploring advanced modeling techniques such as ensemble methods or deep learning architectures
- Evaluating model performance across different market segments and regions to ensure robustness.
- Conducting a more thorough error analysis to identify specific areas for improvement.
- Implementing a feedback loop from the deployed model to continuously improve performance based on real-world data and user interactions.

15 Team Contributions

Task	Members
Data Acquisition	Ahmed Aladdin & Ahmed Hamdy
Data Validation	Ahmed Hamdy & Ahmed Aladdin
Preprocessing	Ahmed Hamdy & Ahmed Aladdin
Feature Engineering & Selection	Asmaa Mohamed & Shehab Khaled
EDA & Visualization	Ahmed Aladdin & Shehab Khaled
Model Development	Asmaa Mohamed & Shehab Khaled
Experiment Tracking	Asmaa Mohamed & Shehab Khaled
Testing	Ahmed Hamdy & Asmaa Mohamed
Results & Evaluation	Asmaa Mohamed & Shehab Khaled
Code Automation & CI	Ahmed Hamdy & Ahmed Aladdin
Deployment & Dashboard	Ahmed Hamdy & Ahmed Aladdin